Name : AKASH SAMPATH

TOPIC : Project 5 :

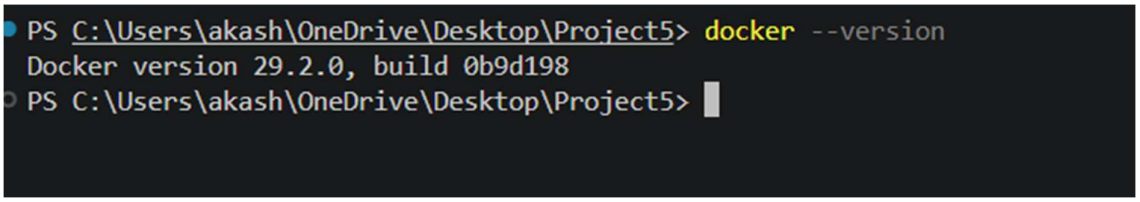
Question :

- Create multiple versions such as V1, V2 of your app
- Build separate docker images
- Push them to Docker Hub or private repo
- Pull and run any version anytime

Check docker installed :

- docker –-version

Output :



Create your app such as v1 and v2 :

Create a folders such as :

- mkdir myapp
- cd myapp

Output :

```
Docker version 29.2.0, build 0b9d198
PS C:\Users\akash\OneDrive\Desktop\Project5> mkdir myapp

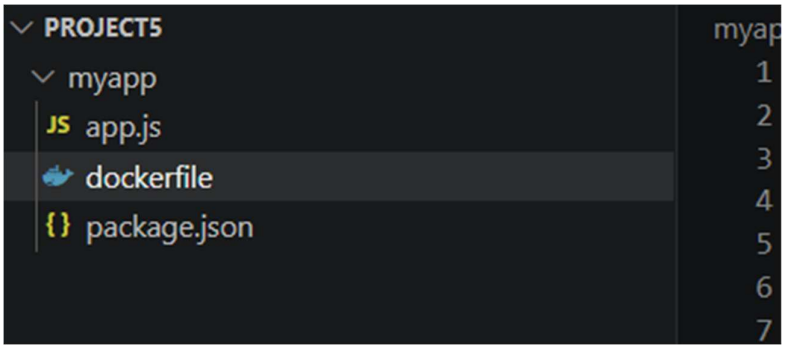

    Directory: C:\Users\akash\OneDrive\Desktop\Project5


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        21-04-2026     22:40                myapp


PS C:\Users\akash\OneDrive\Desktop\Project5> cd myapp
PS C:\Users\akash\OneDrive\Desktop\Project5\myapp>
```

Create the following files for v1 and v2 : app.js , package.json , Dockerfile : And give the codes :

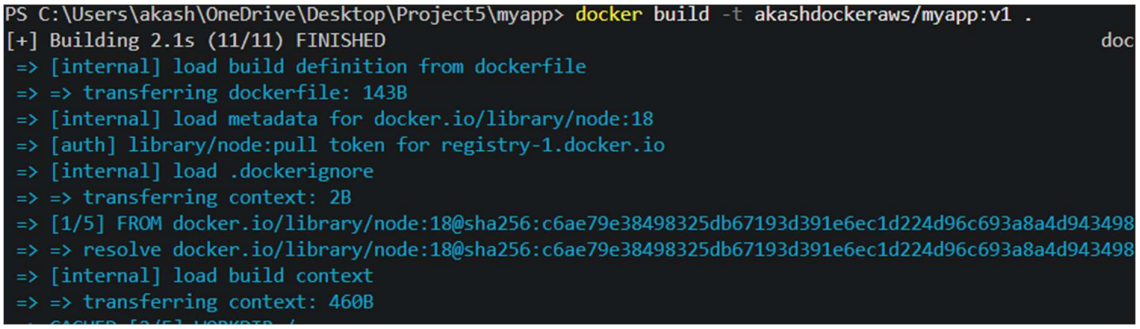
Output :



Build docker image for v1 :

- docker build -t akashdockeraws/myapp:v1 .

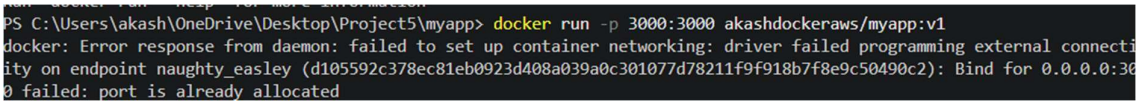
Output :

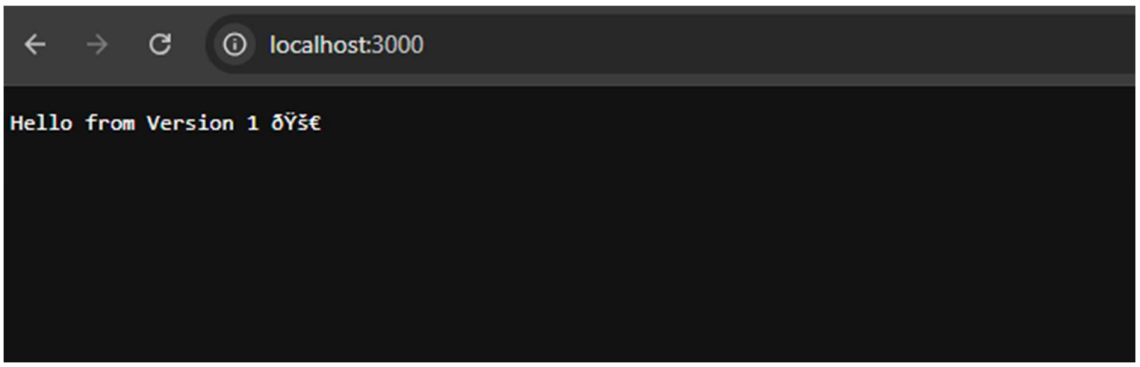


Run v1 Container :

- docker run -p 3000:3000 akashdockeraws /myapp:v1
- localhost:3000

Output :





Create v2 app :

- In app.js replace the code with this : res.end("Hello from Version 2 🔥 ");

Build v2 image :

- docker build -t akashdockeraws/myapp:v2 .

Output :

```
PS C:\Users\akash\OneDrive\Desktop\Project5\myapp> docker build -t akashdockeraws/myapp:v2 .
[+] Building 1.9s (11/11) FINISHED                                                    docker:desktop-lin
 => [internal] load build definition from dockerfile                                                  0.
 => => transferring dockerfile: 143B                                                                  0.
 => [internal] load metadata for docker.io/library/node:18                                            1.
 => [auth] library/node:pull token for registry-1.docker.io                                           0.
 => [internal] load .dockerignore                                                                     0.
 => => transferring context: 2B                                                                       0.
```

Run v2 :

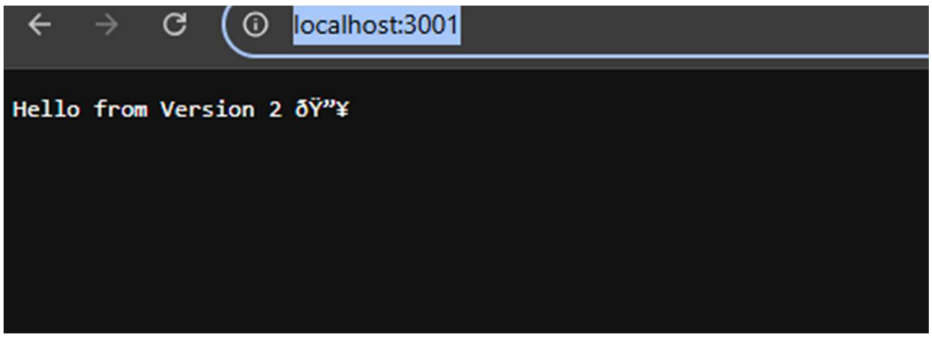
- docker run -p 3001:3000 akashdockeraws/myapp:v2

Output :

```
PS C:\Users\akash\OneDrive\Desktop\Project5\myapp> docker run -p 3001:3000 akashdockeraws/myapp:v2
docker: Error response from daemon: failed to set up container networking: driver failed programming external co
ity on endpoint tender_raman (a5db288c7bf5a4904175875943b7f68ff70e075bc4905edbf326ea0d845112db): Bind for 0.0.0.
failed: port is already allocated
```

- http://localhost:3001/

Output :



Push images to docker hub :

- docker push akashdockeraws/myapp:v1

Output :

```
PS C:\Users\akash\OneDrive\Desktop\Project5\myapp> docker push akashdockeraws/myapp:v1
The push refers to repository [docker.io/akashdockeraws/myapp]
61025853f828: Pushed
481b56a1338b: Pushed
37927ed901b1: Pushed
cda7f44f2bdd: Pushed
461077a72fb7: Pushed
79b2f47ad444: Pushed
3697be50c98b: Pushed
c6b30c3f1696: Pushed
e23f099911d6: Pushed
67a82f46d0d6: Pushed
5497bdfb10b3: Pushed
8ed72933d8d9: Pushed
3e6b9d1a9511: Pushed
```

- **docker push akashdockeraws/myapp:v2**

```
PS C:\Users\akash\OneDrive\Desktop\Project5\myapp> docker push akashdockeraws/myapp:v2
The push refers to repository [docker.io/akashdockeraws/myapp]
9da0b13d9c4a: Pushed
8ed72933d8d9: Layer already exists
37927ed901b1: Layer already exists
3e6b9d1a9511: Layer already exists
cda7f44f2bdd: Layer already exists
461077a72fb7: Layer already exists
61025853f828: Layer already exists
c6b30c3f1696: Layer already exists
3697be50c98b: Layer already exists
5497bdfb10b3: Layer already exists
79b2f47ad444: Layer already exists
1d521f611113: Pushed
e23f099911d6: Layer already exists
v2: digest: sha256:7edfb0752ca29b305d015bb8fbb8c96b9ed967da9076a7eabe36e02136ac1673 size: 856
```

Pull images anytime :

- docker pull akashdockeraws/myapp:v1

Output :

```
PS C:\Users\akash\OneDrive\Desktop\Project5\myapp> docker pull akashdockeraws/myapp:v1
v1: Pulling from akashdockeraws/myapp
Digest: sha256:a3f7e45822a8f86e83ece100576e7d6740aa299999bbaac43f561bf5508a49be
Status: Image is up to date for akashdockeraws/myapp:v1
```

- **docker pull akashdockeraws/myapp:v2**

Output :

```
PS C:\Users\akash\OneDrive\Desktop\Project5\myapp> docker pull akashdockeraws/myapp:v2
v2: Pulling from akashdockeraws/myapp
Digest: sha256:7edfb0752ca29b305d015bb8fbb8c96b9ed967da9076a7eabe36e02136ac1673
Status: Image is up to date for akashdockeraws/myapp:v2
docker.io/akashdockeraws/myapp:v2
```

Run any version anytime :

- docker run -p 3000:3000 akashdockeraws/myapp:v1
- docker run -p 3000:3000 akashdockeraws/myapp:v2

Output :

```
PS C:\Users\akash\OneDrive\Desktop\Project5\myapp> docker run -p 3000:3000 akashdockeraws/myapp:v1
docker: Error response from daemon: failed to set up container networking: driver failed programming external connec
ity on endpoint dazzling_margulis (09f968fb13a24782268222a710b477a5d0f07a17c49ac336157d6eb628ff9ade): Bind for 0.0.0
3000 failed: port is already allocated

Run 'docker run --help' for more information
PS C:\Users\akash\OneDrive\Desktop\Project5\myapp> docker run -p 3000:3000 akashdockeraws/myapp:v2
docker: Error response from daemon: failed to set up container networking: driver failed programming external connec
ity on endpoint keen_carver (4bdf465cdc9926b9ba796509d4752920052fdf5d316a8ecf2e190122d55026e9): Bind for 0.0.0.0:300
ailed: port is already allocated
```

Use private repository :

- **In docker hub create or select a repo created and set it private**
- docker push username/myapp:v1

Output :

